

Dynamic Mode Decomposition (DMD)

Aim: Learn linear models from snapshot data.

~~Recap~~ Given snapshot matrices

$$X = \begin{bmatrix} 1 & & 1 \\ x_1 & \dots & x_m \\ 1 & & 1 \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} 1 & & 1 \\ y_1 & \dots & y_m \\ 1 & & 1 \end{bmatrix},$$

$d \times m$ $d \times m$

where $y_j = \tau x_j \quad j=1, \dots, m,$

DMD constructs the best linear fit

$$(*) \quad \hat{\tau} = \underset{S \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \|Y - SX\|_F.$$

$\Rightarrow$ For low-dimensional systems ($d \ll m$),
(*) is an overdetermined LS problem.

$\Rightarrow$ For high-dimensional systems ($d \gg m$),
(*) is an underdetermined sys. of eqn's.

Q: Which soln should we pick?

A common choice uses the pseudoinverse of X . If the "economy" SVD of X is

$$X = \underbrace{\begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ u_1 & \dots & u_m \\ & & \\ & & 1 \end{bmatrix}}_U \underbrace{\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_m \end{bmatrix}}_\Sigma \underbrace{\begin{bmatrix} -v_1^T \\ \vdots \\ -v_m^T \end{bmatrix}}_{V^T},$$

then the pseudoinverse of X is

$$X^+ = V \Sigma^+ U^T, \text{ where } \Sigma_{jj}^+ = \begin{cases} \sigma_j^{-1}, & \sigma_j \neq 0 \\ 0, & \sigma_j = 0 \end{cases}.$$

The pseudoinverse always provides a minimizer of $(*)$, defined by

$$(**) \quad \hat{\tau} = Y X^+.$$

If $\text{rank}(X) = m$, this is either

$\Rightarrow$ the unique minimizer ($d \geq m$)

$\Rightarrow$ the minimal-norm minimizer ($d \leq m$).

and the pseudoinverse can be written

$$X^\dagger = (X^T X)^{-1} X^T$$

Note that $X^\dagger X$ and $XX^\dagger$ are orthogonal projectors onto the row and column spaces of X , respectively.

"Exact" DMD

The solution (~~44~~) has a key property:

$$\hat{T} = \begin{bmatrix} & \\ & \\ Y & \end{bmatrix} \begin{bmatrix} m \times d \\ \\ \end{bmatrix}$$

$Y \qquad X^\dagger$

so $\text{rank}(\hat{T}) \leq m$. This means we can do, e.g., prediction efficiently:

$$u = \hat{T} v = Y(X^\dagger v).$$

$\Rightarrow$ Mat-vecs scale $\sim dm$ (instead of T^2).

Q: Can we compute the eigenvalues and eigenvectors of T efficiently?

$\Rightarrow$ Standard methods scale $\sim d^3$

Exact DMD uses a clever trick to get the nonzero eigenvalues of T in $\mathcal{O}(m^3)$ and the eigenvectors in $\mathcal{O}(dm^2)$.

Proposition: Given $m \times n$ matrix A and $n \times m$ matrix B , it holds that

- a) the nonzero eigenvalues of AB and BA are identical.
- b) if v is an eigenvector of BA , then $w = Av$ is an eigenvector of AB .

PF Suppose that $BAv = \lambda v$ for $\lambda \neq 0, v \neq 0$.

$\Rightarrow ABv = \lambda Av \Rightarrow ABw = \lambda w$, for $w = Av$.

Note that $w \neq 0$ since $\lambda \neq 0$, otherwise

$$0 = Bw = BA v = \lambda v \neq 0.$$

Similarly, if $ABw = \lambda w$ for $\lambda \neq 0, w \neq 0$

$$\Rightarrow BABw = \lambda Bw \Rightarrow BA v = \lambda v, \text{ for } v = Bw$$

Again, we must have $v \neq 0$.

The Prop. tells us how to find the (nonzero) eigenvalues of $\hat{T}$ by solving an $m \times m$ eigenvalue problem instead of solving a dual eigenvalue problems.

$$\Rightarrow \text{Solve } (X^T Y) v = \lambda v$$

$$\Rightarrow \text{Then } Y X^T w = \lambda w, \quad w = Y v.$$

In practice, DMD uses a slightly different formulation based on the SVD of X . We have that

$$\hat{T} = Y V \Sigma^+ U^T,$$

which has the same nonzero eigenvalues as

$$U^T \hat{T} U = U^T Y V \Sigma^+.$$

If $U^T \hat{T} U v = \lambda v$ and $\lambda \neq 0$, then

$$\hat{T} w = \lambda w, \quad w = Y V \Sigma^+ v.$$

$\Rightarrow$ The matrix $U^T \hat{T} U$ is a block analogue of the Rayleigh quotient.

$\Rightarrow$ In practice, the SVD of X is often truncated further to rank $r < m$

$$X_r = U_r \Sigma_r V_r^T$$

before computing eigenpairs of $U_r^T \hat{T} U_r$.

